

Research Proposal

Working Title

Support Creation Dynamics in Football: A Temporal Framework for Quantifying How Off-Ball Movement Influences Possession Outcomes

Authors

Ryan Inghilterra, Rui [Surname]

Executive Summary

This research proposes a framework for quantifying how off-ball movement influences the quality of support available to a player in possession and how changes in support quality affect future possession outcomes.

While existing football analytics research has focused on pass completion probability, receiver selection, pitch control, and action valuation, relatively little work has examined the temporal process through which support opportunities emerge, improve, deteriorate, or disappear.

Most existing approaches evaluate support at a single moment in time.

This work instead models support as a dynamic process.

The proposed framework introduces a Temporal Support Opportunity Quality (SOQ) measure that incorporates both the current support state and how that state has evolved over recent seconds through player movement.

We then investigate whether improvements in support quality are associated with improved future possession outcomes including possession retention, territorial progression, and threat generation.

Finally, we explore recurring support creation movement archetypes through unsupervised learning.

The resulting framework aims to provide both an interpretable research contribution and practical football applications.

Motivation

Football is fundamentally a game of creating and exploiting options.

When a player is in possession, teammates continuously move to:

- Create passing options
- Create passing lanes
- Increase available space
- Reduce defensive pressure
- Improve progression opportunities

Current analytical approaches primarily answer questions such as:

- Which player is open?
- Which pass is available?
- Which receiver is most likely to be selected?

These approaches evaluate support as a static state.

They do not explicitly quantify how support opportunities are created through movement over time.

The central motivation of this work is to shift the analysis from:

"Who is open?"

to

"How did the support environment evolve to create or remove opportunities for the player in possession?"

Research Questions

RQ1

Can support opportunity quality be quantified using interpretable spatiotemporal features derived from tracking and event data?

RQ2

Can temporal changes in support opportunity quality be measured and linked to future possession outcomes?

RQ3

Do possessions exhibiting stronger support creation dynamics achieve superior retention, progression, and threat outcomes?

RQ4

Can recurring support creation movement patterns be identified through unsupervised learning?

Core Hypothesis

Support opportunities are dynamic.

Off-ball movement continuously changes the support environment available to the player in possession.

Possessions exhibiting positive support creation dynamics should be more likely to:

- Retain possession
- Progress the ball
- Increase future attacking potential

than possessions where support quality remains static or deteriorates.

Proposed Framework

Temporal Support Opportunity Quality (SOQ)

SOQ represents the quality of support available to the player currently in possession.

Importantly, SOQ is not purely frame-based.

It incorporates both:

Instantaneous State

The current support environment.

Temporal Evolution

How the support environment has changed over the recent past.

This allows the framework to distinguish between:

- A player who has been standing open
- A player whose movement recently created a valuable support option

even if both appear identical at the current frame.

Methodology

Step 1: Possession Sequence Identification

Extract possession sequences satisfying:

- Team in possession
- Ball in play
- First two-thirds of pitch
- Minimum possession duration

Possessions will be centered around meaningful football events including:

- Passes
- Turnovers
- Progressive actions
- Possession-ending actions

Rather than enforcing a fixed time window, analysis windows will adapt to possession structure.

Step 2: Temporal SOQ Construction

For each possession frame, calculate a support opportunity quality score using interpretable football features.

Instantaneous Features

Support Availability

- Number of available options
- Support distances
- Support angles

Passing Lane Quality

- Lane openness
- Defender obstruction
- Interception risk

Space Metrics

- Local available space
- Defender density
- Accessible space

Pressure Metrics

- Nearest defender distance
- Pressure intensity estimates

Positional Structure

- Width
 - Depth
 - Balance of support structure
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Temporal Features

Calculated over a preceding temporal window.

Availability Change

- Change in available options
- Change in support distances

Lane Improvement

- Change in passing lane quality
- Newly opened lanes

Space Creation

- Growth in available space
- Defender displacement

Movement Dynamics

- Support displacement
- Velocity
- Acceleration
- Direction changes

Support Stability

- Persistence of support opportunities
 - Volatility of support environment
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Step 3: SOQ Calibration

To avoid arbitrary weighting, SOQ components will be calibrated using interpretable supervised learning techniques.

Potential approaches include:

- Logistic regression
- Elastic Net
- Generalized Additive Models (GAMs)

Target outcomes may include:

- Possession retained after 5 seconds
- Positive progression outcome
- Positive threat generation outcome

The objective is not to maximize predictive performance.

The objective is to learn empirically supported component weights while maintaining interpretability.

Cross-validation and held-out match evaluation will be used to ensure generalization.

Support Creation Dynamics (SCD)

Once SOQ has been estimated, support creation dynamics will be analyzed across possession windows.

Potential metrics include:

Improvement Metrics

- SOQ change
- Peak improvement
- Improvement rate

Stability Metrics

- Time above threshold
- Variance of SOQ
- Persistence of support quality

Opportunity Metrics

- Number of options created
- Number of options lost

These metrics describe how support evolves during possession.

Validation Strategy

Validation is a primary contribution of the paper.

The objective is to determine whether support creation dynamics are associated with meaningful football outcomes.

Validation 1: Possession Retention

Measure:

- Possession retained after 3 seconds
- Possession retained after 5 seconds
- Possession retained after 10 seconds

Hypothesis:

Possessions with stronger support creation dynamics exhibit higher retention rates.

Validation 2: Territorial Progression

Measure:

- Progressive passes
- Progressive carries
- Final-third entries

Hypothesis:

Improved support opportunities facilitate progression.

Validation 3: Threat Generation

Measure future change in:

- Expected Threat (xT)
- Expected Possession Value (EPV)
- Alternative possession value metrics

Hypothesis:

Positive support creation dynamics increase future attacking potential.

Validation 4: Baseline Comparisons

Compare against simpler alternatives including:

- Ball-carrier pressure only
- Pitch location only
- Number of nearby teammates
- Static support measures

Hypothesis:

Temporal support dynamics provide information beyond existing contextual features.

Unsupervised Archetype Discovery

This component is exploratory and secondary to the primary framework.

Using high-support-creation possession windows, derive movement features including:

- Displacement
- Velocity
- Acceleration
- Space creation
- Lane improvement
- Defender separation changes

Apply clustering methods such as:

- HDBSCAN
- K-Means
- Gaussian Mixture Models

to identify recurring support creation behaviors.

Potential archetypes may include:

- Check-to-ball support
- Lateral escape support
- Progressive support runs
- Width-creating support
- Third-man support movements

The goal is descriptive understanding rather than prediction.

Dataset

Primary Dataset:

PFF FC FIFA World Cup 2022 Tracking and Event Data

Advantages:

- Tournament-wide coverage
- Tracking and event synchronization
- Large sample size
- Suitable for publication-quality analysis

Secondary Dataset (Optional):

SkillCorner Open Data

Potentially used for prototyping, replication, or external validation.

Expected Contributions

Contribution 1

A temporal framework for quantifying support opportunity quality.

Contribution 2

A methodology for measuring support creation dynamics during possession.

Contribution 3

Empirical evidence linking support creation to future possession outcomes.

Contribution 4

A taxonomy of recurring support creation movement patterns.

Novelty

Most existing football analytics work evaluates:

- Pass quality

- Pass completion probability
- Receiver selection
- Pitch control
- Action valuation

This work instead focuses on:

How support opportunities evolve through off-ball movement before possession outcomes occur.

The central contribution is not identifying which options exist at a moment in time.

The contribution is quantifying how support opportunities are created, strengthened, weakened, or lost during possession.

This shifts the analysis from static opportunity evaluation toward the temporal dynamics of support creation.